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"""
CONN Capacity Test: Scaling to N=2048
Parameters:  $\lambda=0.5$  (annealed),  $\eta=0.02$ , steps=400, jitter=0.5, noise=0.30
Objective: Test if Improvement Ratio reaches 1.70x - 1.80x
"""

import numpy as np

def generate_patterns(M, N, seed):
    np.random.seed(seed)
    return np.random.choice([0, np.pi], size=(M, N))

def bitflip_plus_jitter(pattern, noise_level, jitter_std, rng):
    noisy = pattern.copy()
    mask = rng.random(len(pattern)) < noise_level
    noisy[mask] = (noisy[mask] + np.pi) % (2 * np.pi)
    noisy = noisy + rng.normal(0, jitter_std, len(pattern))
    return np.mod(noisy, 2 * np.pi)

def compute_overlap(retrieved, target):
    r = np.where(np.cos(retrieved) >= 0, 0.0, np.pi)
    t = np.where(np.cos(target) >= 0, 0.0, np.pi)
    return np.sum(np.abs(np.cos(r) - np.cos(t)) < 0.5) / len(target) * 100

def hebbian_weights(patterns):
    xi = np.cos(patterns)
    J = (xi.T @ xi) / patterns.shape[1]
    np.fill_diagonal(J, 0)
    return J

def hopfield_recall(J, noisy, max_steps=200):
    s = np.cos(noisy).copy()
    for _ in range(max_steps):
        s_new = np.sign(J @ s)
        s_new[s_new == 0] = 1
        if np.array_equal(s_new, s):
            break
        s = s_new
    return np.where(s > 0, 0.0, np.pi)

def conn_annealed(J, noisy, lam, eta_phi, steps):
    N = len(noisy)
    phi = noisy.copy()
    A = np.ones(N)
    for step in range(steps):
        t = step / steps
        lam_t = lam * t
        s, c = np.sin(phi), np.cos(phi)
        # Core CONN Dynamics
        coupling = (J @ (A * s)) * c - (J @ (A * c)) * s
        coherence = -2 * lam_t * A**2 * s * c
        dphi = A * coupling + coherence
        phi = (phi + eta_phi * dphi) % (2 * np.pi)

        # Amplitude Annealing
        dA = -2 * lam_t * A * s**2 - 1.0 * (A - 1)
        A = np.clip(A + 0.03 * dA, 0.01, 2.0)
    return phi

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# MAIN EXPERIMENT - N=2048
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print("=" * 90)
print("CONN vs Hopfield Capacity Test - Hyper Scaling (N=2048)")
print("Parameters:  $\lambda=0.5$  (annealed),  $\eta=0.02$ , steps=400, jitter=0.5,
noise=0.30")
print("=" * 90)

noise = 0.30
jitter = 0.5
trials = 30
test_patterns_per_trial = 3

for N in [2048]:
    print(f"\n{'='*80}")
    print(f"                                N = {N}")
    print(f{'='*80})
    print(f"    {'M':>4s} {'Hopfield':>10s} {'CONN':>8s} {' $\Delta$ ':>8s}
{'hop_std':>8s} {'conn_std':>8s}")
    print(f"    {'-'*62}")

    hop_cap = 0
    conn_cap = 0
    max_M = int(0.35 * N)    # Testing up to M=716

    for M in range(1, max_M + 1):
        hop_ov = []
        conn_ov = []

        for trial in range(trials):
            patterns = generate_patterns(M, N, seed=trial)
            J = hebbian_weights(patterns)

            for mu in range(min(M, test_patterns_per_trial)):
                rng = np.random.RandomState(trial * 1000 + mu)
                noisy = bitflip_plus_jitter(patterns[mu], noise, jitter,
rng)

                # Baseline Recall
                ret_h = hopfield_recall(J, noisy)
                hop_ov.append(compute_overlap(ret_h, patterns[mu]))

                # CONN Recall
                ret_c = conn_annealed(J, noisy, 0.5, 0.02, 400)
                conn_ov.append(compute_overlap(ret_c, patterns[mu]))

        h_mean = np.mean(hop_ov)
        c_mean = np.mean(conn_ov)
        h_std = np.std(hop_ov)
        c_std = np.std(conn_ov)

        # Capture threshold capacity
        if h_mean >= 80: hop_cap = M
        if c_mean >= 80: conn_cap = M

        # Periodic Reporting
        if (M <= 5 or M % 20 == 0 or
            abs(h_mean - 80) < 3 or abs(c_mean - 80) < 3):
            print(f"    {M:4d} {h_mean:9.1f}% {c_mean:7.1f}% {c_mean-
h_mean:+7.1f}% {h_std:7.1f}% {c_std:7.1f}%")

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# Termination: stop once both models have effectively failed
if h_mean < 55 and c_mean < 55 and M > 250:
    break

ratio = conn_cap / hop_cap if hop_cap > 0 else 0
print(f"\n [Hopfield capacity: M={hop_cap:3d} ( $\alpha$ ={{hop_cap/N:.3f}})"]
print(f" | Hopfield capacity: M={hop_cap:3d} ( $\alpha$ ={{hop_cap/N:.3f}})
|")
print(f" | CONN capacity:      M={{conn_cap:3d} ( $\alpha$ ={{conn_cap/N:.3f}})
|")
print(f" | Improvement ratio: {{ratio:.2f}}×
print(f" | Improvement ratio: {{ratio:.2f}}×")

print("\n" + "="*90)
print("Hyper-scaling run completed.")

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RESULTS OF THE COLAB RUN:

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CONN vs Hopfield Capacity Test - Hyper Scaling (N=2048)
Parameters:  $\lambda$ =0.5 (annealed),  $\eta$ =0.02, steps=400, jitter=0.5, noise=0.30
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N = 2048
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M	Hopfield	CONN	Δ	hop_std	conn_std
1	100.0%	90.5%	-9.5%	0.0%	2.1%
2	100.0%	90.6%	-9.4%	0.0%	2.1%
3	100.0%	90.8%	-9.2%	0.0%	1.7%
4	100.0%	90.9%	-9.1%	0.0%	1.8%
5	100.0%	91.0%	-9.0%	0.0%	1.7%
20	100.0%	91.8%	-8.2%	0.0%	1.5%
40	100.0%	92.2%	-7.8%	0.0%	1.4%
60	100.0%	92.1%	-7.9%	0.0%	1.4%
80	100.0%	91.7%	-8.3%	0.0%	1.4%
100	100.0%	91.2%	-8.8%	0.0%	1.4%

120	100.0%	90.5%	-9.5%	0.0%	1.5%
140	100.0%	89.8%	-10.2%	0.0%	1.6%
160	100.0%	89.1%	-10.9%	0.0%	1.8%
180	100.0%	88.4%	-11.6%	0.1%	2.0%
200	98.6%	87.5%	-11.1%	5.6%	1.9%
220	92.3%	86.6%	-5.7%	13.1%	1.9%
233	80.9%	86.0%	+5.1%	16.0%	1.8%
234	81.9%	85.9%	+4.0%	15.4%	1.9%
235	79.5%	85.9%	+6.4%	15.8%	1.9%
236	80.0%	85.9%	+5.8%	15.5%	1.9%
237	79.4%	85.8%	+6.5%	15.7%	1.9%
238	80.4%	85.8%	+5.4%	15.2%	1.9%
239	78.5%	85.8%	+7.3%	15.5%	2.0%
240	77.7%	85.7%	+8.0%	15.2%	1.9%
241	77.7%	85.7%	+8.0%	14.9%	1.9%
242	77.0%	85.6%	+8.6%	14.8%	2.0%
260	70.2%	85.0%	+14.8%	10.2%	2.0%
280	66.1%	84.1%	+18.0%	5.7%	2.1%
300	64.9%	83.3%	+18.4%	3.7%	2.1%
306	64.9%	83.0%	+18.1%	3.6%	2.0%
307	64.5%	82.9%	+18.4%	3.9%	2.0%
308	65.1%	82.9%	+17.8%	4.3%	2.1%
309	64.4%	82.9%	+18.5%	3.9%	2.0%
310	64.7%	82.9%	+18.1%	3.5%	2.0%
311	65.0%	82.8%	+17.7%	3.5%	2.0%
312	65.4%	82.7%	+17.3%	4.3%	1.9%
313	65.2%	82.7%	+17.5%	4.3%	2.0%
314	64.3%	82.7%	+18.4%	3.6%	2.0%
315	64.5%	82.6%	+18.1%	3.1%	2.0%
316	64.4%	82.6%	+18.2%	3.3%	2.0%
317	64.4%	82.5%	+18.1%	3.3%	2.0%
318	64.4%	82.5%	+18.1%	3.4%	2.0%
319	64.3%	82.5%	+18.2%	3.5%	1.9%
320	64.4%	82.5%	+18.1%	3.4%	2.0%
321	64.5%	82.5%	+18.0%	3.9%	2.0%
322	64.6%	82.4%	+17.8%	3.7%	1.9%
323	64.4%	82.4%	+18.1%	4.0%	1.9%
324	63.9%	82.4%	+18.5%	3.2%	1.9%
325	64.0%	82.3%	+18.3%	3.5%	1.9%
326	64.3%	82.3%	+18.1%	3.4%	1.9%
327	64.1%	82.3%	+18.2%	3.5%	1.9%
328	64.3%	82.2%	+17.9%	3.7%	2.0%
329	64.3%	82.2%	+17.8%	3.7%	2.0%
330	64.2%	82.1%	+17.9%	3.9%	1.9%
331	64.7%	82.1%	+17.4%	3.5%	1.9%
332	64.1%	82.0%	+17.9%	3.6%	1.9%
333	64.0%	82.0%	+18.0%	3.6%	1.9%
334	63.9%	82.0%	+18.0%	3.2%	1.8%
335	64.5%	81.9%	+17.4%	3.6%	1.8%
336	64.3%	81.9%	+17.6%	3.1%	1.8%
337	64.5%	81.9%	+17.3%	3.2%	1.8%
338	64.4%	81.8%	+17.4%	3.1%	1.9%
339	64.0%	81.7%	+17.7%	3.0%	1.9%
340	64.2%	81.7%	+17.5%	3.5%	1.9%
341	64.3%	81.6%	+17.4%	3.2%	1.9%
342	63.9%	81.6%	+17.7%	3.4%	1.9%
343	64.0%	81.5%	+17.6%	3.2%	1.9%
344	64.3%	81.5%	+17.2%	3.4%	1.9%
345	63.7%	81.5%	+17.8%	3.3%	1.9%
346	64.2%	81.4%	+17.2%	3.5%	1.9%
347	64.1%	81.4%	+17.3%	3.0%	1.9%
348	64.0%	81.3%	+17.3%	2.9%	1.9%
349	63.8%	81.2%	+17.4%	3.2%	1.9%

350	63.8%	81.2%	+17.4%	3.6%	1.9%
351	63.6%	81.2%	+17.6%	3.4%	1.9%
352	64.0%	81.2%	+17.2%	3.1%	1.9%
353	63.6%	81.2%	+17.5%	3.4%	1.9%
354	64.2%	81.1%	+16.9%	3.3%	1.9%
355	63.8%	81.1%	+17.2%	3.3%	1.9%
356	63.5%	81.0%	+17.6%	3.7%	1.9%
357	63.7%	81.0%	+17.4%	3.6%	1.9%
358	63.7%	81.0%	+17.3%	3.3%	1.9%
359	63.8%	81.0%	+17.2%	3.6%	1.9%
360	63.6%	81.0%	+17.3%	3.4%	1.9%
361	63.5%	80.9%	+17.4%	3.0%	1.9%
362	63.7%	80.9%	+17.2%	3.2%	2.0%
363	63.5%	80.8%	+17.3%	3.3%	1.9%
364	63.7%	80.8%	+17.1%	3.2%	2.0%
365	63.4%	80.7%	+17.3%	3.0%	2.0%
366	63.9%	80.7%	+16.8%	3.2%	1.9%
367	63.2%	80.6%	+17.4%	3.2%	2.0%
368	63.1%	80.6%	+17.6%	3.2%	1.9%
369	63.3%	80.6%	+17.3%	2.8%	1.9%
370	63.6%	80.5%	+16.8%	3.2%	2.0%
371	63.3%	80.5%	+17.2%	3.4%	1.9%
372	63.6%	80.4%	+16.8%	3.3%	1.9%
373	63.7%	80.5%	+16.7%	2.9%	1.9%
374	63.7%	80.4%	+16.7%	3.3%	1.9%
375	63.7%	80.4%	+16.7%	3.4%	1.9%
376	63.6%	80.3%	+16.7%	2.9%	1.8%
377	63.2%	80.3%	+17.1%	2.9%	1.9%
378	62.9%	80.2%	+17.4%	3.5%	1.9%
379	63.4%	80.2%	+16.9%	3.2%	1.9%
380	63.5%	80.2%	+16.7%	3.2%	1.9%
381	63.4%	80.2%	+16.8%	3.2%	1.8%
382	63.3%	80.2%	+16.8%	3.3%	1.9%
383	63.0%	80.1%	+17.1%	3.4%	1.8%
384	63.0%	80.1%	+17.1%	3.3%	1.8%
385	63.0%	80.1%	+17.2%	3.0%	1.8%
386	63.2%	80.1%	+16.9%	3.2%	1.9%
387	62.6%	80.0%	+17.4%	3.2%	1.8%
388	63.2%	80.0%	+16.8%	3.2%	1.8%
389	62.6%	79.9%	+17.3%	3.1%	1.8%
390	62.9%	80.0%	+17.0%	3.0%	1.8%
391	62.6%	79.9%	+17.3%	3.3%	1.8%
392	62.8%	79.9%	+17.1%	3.0%	1.8%
393	62.9%	79.8%	+16.9%	3.3%	1.8%
394	62.9%	79.8%	+16.8%	3.0%	1.8%
395	62.2%	79.7%	+17.5%	3.0%	1.8%
396	62.6%	79.7%	+17.1%	3.1%	1.8%
397	62.7%	79.7%	+17.0%	2.8%	1.8%
398	62.5%	79.7%	+17.1%	2.8%	1.8%
399	62.4%	79.6%	+17.2%	3.0%	1.8%
400	62.6%	79.6%	+17.0%	2.9%	1.8%
401	62.7%	79.6%	+16.8%	3.2%	1.8%
402	62.2%	79.5%	+17.4%	3.3%	1.8%
403	62.4%	79.5%	+17.1%	3.1%	1.8%
404	62.5%	79.5%	+16.9%	3.1%	1.8%
405	62.6%	79.5%	+16.9%	2.9%	1.7%
406	62.3%	79.4%	+17.2%	3.0%	1.8%
407	62.2%	79.4%	+17.3%	3.0%	1.8%
408	62.2%	79.4%	+17.2%	2.8%	1.8%
409	62.8%	79.4%	+16.6%	2.9%	1.8%
410	62.5%	79.3%	+16.9%	3.0%	1.8%
411	62.6%	79.3%	+16.7%	2.8%	1.9%
412	62.6%	79.3%	+16.7%	2.9%	1.9%

413	62.4%	79.2%	+16.8%	2.8%	1.8%
414	62.6%	79.2%	+16.6%	2.9%	1.8%
415	62.6%	79.2%	+16.6%	3.1%	1.9%
416	62.4%	79.1%	+16.8%	2.9%	1.9%
417	62.5%	79.1%	+16.7%	2.7%	1.8%
418	62.7%	79.2%	+16.5%	2.6%	1.9%
419	62.7%	79.2%	+16.5%	3.1%	1.8%
420	62.8%	79.1%	+16.3%	2.9%	1.8%
421	62.5%	79.0%	+16.5%	3.0%	1.9%
422	62.7%	79.0%	+16.2%	3.1%	1.9%
423	62.2%	79.0%	+16.8%	3.0%	1.8%
424	62.3%	78.9%	+16.7%	3.2%	1.8%
425	62.3%	78.8%	+16.5%	3.1%	1.8%
426	62.6%	78.9%	+16.3%	3.3%	1.8%
427	62.5%	78.8%	+16.3%	2.8%	1.8%
428	61.9%	78.8%	+16.8%	3.1%	1.9%
429	62.4%	78.7%	+16.3%	2.9%	1.9%
430	62.3%	78.7%	+16.4%	2.8%	1.9%
431	62.5%	78.6%	+16.2%	2.8%	1.9%
432	62.1%	78.6%	+16.5%	2.9%	1.9%
433	61.9%	78.7%	+16.7%	2.9%	1.9%
434	62.1%	78.6%	+16.5%	2.9%	1.9%
435	62.3%	78.6%	+16.4%	2.8%	1.9%
436	62.0%	78.6%	+16.6%	2.9%	1.9%
437	61.9%	78.6%	+16.7%	3.0%	1.9%
438	62.1%	78.5%	+16.4%	2.7%	1.9%
439	62.1%	78.5%	+16.4%	2.9%	1.9%
440	61.7%	78.5%	+16.8%	2.7%	2.0%
441	61.9%	78.5%	+16.6%	3.0%	2.0%
442	61.7%	78.4%	+16.7%	2.7%	2.0%
443	62.0%	78.4%	+16.3%	2.8%	2.0%
444	61.9%	78.4%	+16.5%	2.7%	2.0%
445	62.0%	78.4%	+16.4%	2.7%	2.0%
446	62.0%	78.3%	+16.3%	2.9%	2.0%
447	61.5%	78.3%	+16.8%	2.8%	2.1%
448	62.0%	78.2%	+16.3%	3.0%	2.0%
449	61.4%	78.2%	+16.8%	2.8%	2.0%
450	61.6%	78.1%	+16.5%	2.8%	2.0%
451	61.5%	78.1%	+16.7%	2.8%	2.0%
452	61.5%	78.1%	+16.6%	2.8%	2.0%
453	61.5%	78.1%	+16.6%	2.9%	2.1%
454	61.6%	78.1%	+16.4%	2.4%	2.0%
455	61.7%	78.1%	+16.4%	2.8%	2.0%
456	61.5%	78.1%	+16.6%	2.7%	2.0%
457	61.9%	78.0%	+16.1%	2.7%	2.0%
458	61.3%	78.0%	+16.7%	2.8%	1.9%
459	61.5%	77.9%	+16.4%	2.9%	1.9%
460	61.8%	77.9%	+16.2%	2.6%	1.9%
461	61.4%	77.9%	+16.5%	2.5%	2.0%
462	61.6%	77.9%	+16.3%	2.7%	2.0%
463	61.4%	77.8%	+16.4%	2.5%	2.0%
464	61.5%	77.8%	+16.3%	2.6%	1.9%
465	61.7%	77.8%	+16.1%	2.8%	1.9%
466	61.7%	77.8%	+16.1%	2.5%	1.9%
467	61.9%	77.7%	+15.9%	2.7%	1.9%
468	61.6%	77.7%	+16.0%	2.6%	1.9%
469	61.5%	77.7%	+16.1%	2.7%	1.9%
470	61.6%	77.6%	+16.0%	2.5%	2.0%
471	61.2%	77.6%	+16.4%	2.7%	1.9%
472	61.6%	77.6%	+15.9%	2.8%	1.9%
473	61.4%	77.5%	+16.1%	2.4%	1.9%
474	61.3%	77.5%	+16.2%	2.9%	1.9%
475	61.3%	77.4%	+16.2%	2.8%	1.9%

476	61.5%	77.4%	+16.0%	2.4%	1.9%
477	61.2%	77.5%	+16.2%	2.8%	1.9%
478	61.4%	77.4%	+16.0%	2.8%	1.9%
479	61.6%	77.4%	+15.8%	2.6%	1.9%
480	61.2%	77.4%	+16.2%	2.8%	1.9%
481	61.5%	77.4%	+15.9%	2.3%	1.9%
482	61.2%	77.3%	+16.2%	2.5%	1.9%
483	61.1%	77.3%	+16.2%	2.6%	1.9%
484	61.2%	77.3%	+16.1%	2.7%	1.9%
485	61.4%	77.2%	+15.9%	2.9%	1.9%
486	61.4%	77.3%	+15.9%	2.6%	1.9%
487	61.5%	77.2%	+15.7%	2.9%	1.9%
488	61.1%	77.2%	+16.1%	3.0%	1.9%
489	61.4%	77.1%	+15.7%	2.5%	1.9%
490	61.3%	77.0%	+15.8%	2.8%	2.0%
491	61.5%	77.0%	+15.5%	2.5%	2.0%